

behaviour? Based on pure logic alone, self-sacrifice is a very bad choice.

Now, reimagine the same scenario with a car driven by AI based software. Since you are the owner of the car, should the AI software be trained so that your safety is the topmost priority, even to the point of utterly disregarding the safety of your co-passengers. It is easy to see that a situation where all the cars in the world are driven by such software will lead to utter chaos. Now, further assume the AI system has the additional information that the passenger riding the self-driving car is suffering from a terminal illness. With this additional information, it is logical (only to a mathematical machine, not to us, flesh and blood humans) to sacrifice the passenger for the pedestrian. I don't want to indulge any more in this thought exercise but it will be rewarding if you could spend some time to think about the ever more complicated scenarios that may arise when decisions are made by logic-oriented machines instead of hot-blooded humans.

There are many books and articles that deal with the political, social, and ethical aspects of AI when it becomes fully operational. But I think for us mere mortals and computer engineers it would be overkill to read all of them. However, since the social relevance of AI is so important, we cannot shelve the issue that easily. To be aware of the sociopolitical aspects of AI, I suggest you watch a few movies to easily understand how AI (strong AI for that matter) may impact all of us. The masterpiece by Stanley Kubrick, '2001: A Space Odyssey', is one of the first movies to depict how a superior intelligent being could look down upon us humans. This movie is one among the many in which AI decides that humanity is the biggest threat to the world and decides to destroy humans. Indeed, there are quite a number of movies where this plot is explored. But that is not the extent of

it. There is a movie called 'Artificial Intelligence' by the great maestro Steven Spielberg himself, which explores how a machine powered with AI interacts with a human being. Another movie called 'Ex Machina' elaborates on this thread and deals with machines capable of artificial general intelligence. In my opinion these are all must-watch movies to understand the impact of AI.

As a final thought exercise, just contemplate roads where every car manufacturing company has its own rules and AI for self-driving cars. That will lead to complete pandemonium.

An introduction to TensorFlow

Now let us learn about TensorFlow, a heavyweight among libraries used for developing AI and machine learning based applications. Notice that, in addition to Python, TensorFlow can also be used with programming languages like C++, Java, JavaScript, etc. TensorFlow is a free and open source library licensed under Apache License 2.0, and was developed by the Google Brain team. But before we proceed any further, let us try to understand what a tensor is. If you are familiar with physics, the term tensor may not be new to you. But even if you are not familiar with the term tensor, no need to worry. Think of tensors as multidimensional arrays. Of course, this is an oversimplification. However, this simplified notion of tensors is sufficient for the time being. TensorFlow can operate on top of multidimensional arrays offered by NumPy.

First, let us install TensorFlow in JupyterLab. The kind of installation required depends on whether your system has a GPU or not. But, what is a GPU (graphics processing unit)? GPU is an electronic circuit that uses parallel processing to accelerate the speed of image processing. Though often used by gamers and designers, GPUs are an essential hardware requirement while developing AI and

machine learning based applications. Unfortunately, not every GPU is compatible with TensorFlow. You need to have an NVIDIA GPU to install the GPU version of TensorFlow. Further, you need to install a parallel computing platform called CUDA (compute unified device architecture) in your system. If your system satisfies these requirements, then execute the command '`pip install tensorflow-gpu`' on JupyterLab to install the GPU version of TensorFlow. If there is something wrong with the GPU configuration of your system, you will get the error message, '`CUDA_ERROR_NO_DEVICE: no CUDA-capable device is detected`', when you try to use TensorFlow. If you get this error, please uninstall the GPU version of TensorFlow with the following command, '`pip uninstall tensorflow-gpu`'. Later, please install the CPU version of TensorFlow by executing the command, '`pip install tensorflow`', on JupyterLab. Now, we are ready to use TensorFlow. Notice that, for the time being, we restrict our discussion to CPUs and TensorFlow.

Now, let us run our first Python code powered with TensorFlow. Figure 1 shows a simple Python script and its output when executed on JupyterLab. The first two lines of code import the libraries NumPy and TensorFlow into our Python script. As a side note, if you want to display line numbers in your Jupyter Notebook cells then click on 'View>Show Line Numbers'. Line 3 creates an array named `arr` with three elements using NumPy. The line of code '`ten = tf.multiply(arr, 3)`' multiplies every element of the array `arr` with the number 3, and the result is stored in a variable named `ten`. Lines 5 and 6 print the types of variables `arr` and `ten`, respectively. From the output of the code you can see that `arr` and `ten` are of different types. Finally, line 7 prints the value of the variable `ten`. Notice that the shape of `ten` is the same as that of array `arr`. Also notice that